Supplementary Material
When More Evidence Hurts: Publication-Bias Drift and Principled Stopping for Biomedical Causal Search

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This supplement provides formal derivations, graph-feature definitions, training details, calibration analysis, and robustness checks accompanying the main paper. Equations and figures are numbered locally.

I. Formal Derivations

A. Vote-Counting Drift

Let $X_i \sim \mathrm{Bernoulli}(p)$ be independent and identically distributed, with $p = \frac{1}{2} + b$, $0 < b < \frac{1}{2}$, and let $\hat{p}_n^+ = n^{-1}\sum_{i=1}^n X_i$. On a true NoEffect query, the vote-counting rule predicts Beneficial if $\hat{p}_n^+ > \frac{1}{2}$. Its false-positive risk is

$$D(n) = P\left(\sum_{i=1}^n X_i > \frac{n}{2}\right).$$

By the strong law of large numbers, $\hat{p}_n^+ \to p > \frac{1}{2}$ almost surely. Thus the error indicator tends to one almost surely, and bounded convergence gives $D(n) \to 1$.

For a large-sample normal approximation, set $\sigma^2 = p(1-p)$. The standardised fraction $(\hat{p}_n^+ - p)/(\sigma/\sqrt{n})$ converges in distribution to a standard normal variable. Applying the normal approximation gives

$$D(n) \approx 1 - \Phi\left(\frac{\frac{1}{2} - p}{\sigma/\sqrt{n}}\right) = \Phi\left(\frac{b\sqrt{n}}{\sigma}\right).$$

The normal approximation gives a strictly increasing large-sample trend. The exact binomial risk is

$$D(n) = \sum_{j=\lfloor n/2 \rfloor + 1}^{n} \binom{n}{j} p^j (1-p)^{n-j}$$

with a discrete threshold that changes with parity. The law of large numbers establishes the limit, while the normal approximation describes the smooth large-sample trend around these finite-sample fluctuations. Here n counts reports and t counts retrieval iterations. An iteration can contribute several reports, so batch size determines how quickly the report count grows with retrieval depth.

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B. Noisy-OR Drift

Suppose each report supports one of two edges for the same entity pair: a positive edge with confidence $s_+ \in (0,1)$ or a neutral edge with confidence $s_0 \in (0,1)$. With $N_n^+ = \sum_i X_i$, noisy-OR gives

$$W_+ = 1 - (1-s_+)^{N_n^+}, \qquad W_0 = 1 - (1-s_0)^{n - N_n^+}.$$

Write $a_+ = -\log(1-s_+)$ and $a_0 = -\log(1-s_0)$. Since the complements are positive,

$$\begin{aligned} W_+ > W_0 &\iff (1-s_+)^{N_n^+} < (1-s_0)^{n-N_n^+} \\ &\iff N_n^+ a_+ > (n - N_n^+) a_0 \\ &\iff \hat{p}_n^+ > \tau, \qquad \tau = \frac{a_0}{a_+ + a_0}. \end{aligned}$$

If $p > \tau$, the probability of selecting the positive branch tends to one by the same law-of-large-numbers argument. The large-sample risk approximation is $\Phi((p-\tau)\sqrt{n}/\sqrt{p(1-p)})$. If $s_+ = s_0$, then $\tau = \frac{1}{2}$, and both rules agree at every finite depth, including ties. A higher positive confidence lowers τ, illustrating how asymmetric extraction can favour the positive branch.

For two routes of the same length, multiplication by their common hop discount and exponentiation by a common $\beta > 0$ preserve score ordering. The two-branch formula isolates the contribution of report frequency and extraction confidence. In the full graph, direct and intermediate routes contribute jointly to three label scores, and a leading-label margin determines whether the agent makes a conclusion. Correlated support and changing route portfolios offer further extensions of the analysis; the simulation examines drift under several such perturbations.

C. KL–Energy Bound

The main algorithm monitors posterior change and supplies uncertainty and low-confidence statistics to the PRM. A three-term diagnostic functional relates these signals. For a fixed label L, write

$$\mathcal{D}_t(L) = -\log q_t(L) + \lambda_U U_t + \lambda_V V_t,$$

with non-negative weights. The main paper defines U_t as the mean polarity entropy of distinct route-covered entity pairs and V_t as the total low-confidence gap over distinct route edges. If all posterior probabilities are at least $\varepsilon_0 > 0$, and $|U_t - U_{t-1}| \le \varepsilon_U$, $|V_t - V_{t-1}| \le \varepsilon_V$, then

$$|\mathcal{D}_t(L) - \mathcal{D}_{t-1}(L)| \le \frac{1}{\varepsilon_0}\sqrt{\frac{1}{2}D_{\mathrm{KL}}(q_t\|q_{t-1})} + \lambda_U\varepsilon_U + \lambda_V\varepsilon_V. \tag{1}$$

To prove this, use Pinsker's inequality [1],

$$\mathrm{TV}(q_t, q_{t-1}) \le \sqrt{\tfrac{1}{2}D_{\mathrm{KL}}(q_t\|q_{t-1})},$$

and the $1/\varepsilon_0$-Lipschitz property of $-\log x$ on $[\varepsilon_0, 1]$. For any fixed label,

$$|q_t(L) - q_{t-1}(L)| \le \mathrm{TV}(q_t, q_{t-1}).$$

The triangle inequality then yields the bound. The same argument controls the change in $-\log \max_L q_t(L)$ because $|\max_L q_t(L) - \max_L q_{t-1}(L)| \le \mathrm{TV}(q_t, q_{t-1})$.

The bound quantifies energy stability when both the posterior and graph statistics vary slowly. The square-root KL term controls the log-posterior contribution, while the two graph terms retain their own variation bounds. This interpretation motivates observing posterior change and graph quality together during retrieval.

D. Bradley–Terry Reward Ranking

Let $c(\phi) = P(\text{correct} \mid \phi) \in (0,1)$ and $u(\phi) = \log[c(\phi)/(1-c(\phi))]$. Assume that population preference probabilities obey the Bradley–Terry model

$$P(i \succ j \mid \phi_i, \phi_j) = \sigma(u(\phi_i) - u(\phi_j)),$$

and that the reward function class can realise u. For the unmodified population pairwise cross-entropy, each pair's conditional loss is minimised when the predicted preference probability matches the specified probability. Strict monotonicity of the logistic function gives

$$r^*(\phi_i) - r^*(\phi_j) = u(\phi_i) - u(\phi_j).$$

On a connected component of the comparison support, this implies $r^*(\phi) = u(\phi) + c_0$. Hence, within that component, reward ranking agrees with correctness-probability ranking.

This derivation gives the ranking interpretation of the specified population preference model. The deployed MLP learns from finite, same-query preference pairs using the margin loss below; held-out pairwise accuracy measures the resulting ranking quality. During retrieval, the agent tracks the running maximum reward and stops at the current state after a decline or a reward plateau. The main ablation evaluates how this online use of the reward shortens retrieval alongside KL convergence.

II. Implementation Details

A. Graph Features

The 20-dimensional feature vector comprises six groups:

Path structure (6): Direct-path count, maximum direct-path score, two-hop count, maximum two-hop score, best overall path score, and gap between best and second-best.

Conflict (2): Fraction of entity pairs with contradictory-polarity edges and polarity entropy.

Coverage (4): Source degree, target degree, common neighbours, and new-edge ratio.

Saturation (2): Path-discovery rate and marginal confidence gain.

Global statistics (4): Total evidence count, mean edge confidence, graph density, and publication recency score.

Diagnostics (2): Mean entity-pair polarity entropy and summed low-confidence gap.

For the diagnostic entropy, evidence counts are grouped by positive, negative, and neutral polarity within an entity pair and normalised to a three-class distribution. Each distinct pair covered by direct and two-hop routes is counted once. For the low-confidence statistic, each route-covered edge contributes $\max(0, 0.3 - w_e)$ once. Entropy is averaged over pairs, whereas gaps are summed over edges; their normalised PRM inputs are $U_t/1.1$ and $V_t/2$. Together, these features describe disagreement and weak support in the routes used for the current conclusion.

B. PRM Training and Stopping

The ReLU MLP has architecture $[20 \to 128 \to 64 \to 32 \to 1]$. The training loss on preferred and dispreferred states is

$$\mathcal{L} = -\log \sigma(r(\phi^+) - r(\phi^-) - m), \qquad m = 0.1.$$

The reported training set contains 2,998 pairs from training-split trajectories, with 479 validation pairs. Preferences are supervised using Cochrane correctness labels and oracle-correct states. Optimisation uses Adam, learning rate 10^{-3}, batch size 32, weight decay 10^{-4}, at most 100 epochs, and early-stopping patience 20. Validation ranking accuracy is 88.1%; the reported preferred–dispreferred reward margin is 2.53 ± 2.16.

Frozen thresholds are $\delta = 0.01$ for KL, $\alpha = 0.3$ for decline from the running reward maximum, and $\alpha_c = 0.1$ for the range of four successive rewards. The OR policy stops when any trigger fires and returns the current graph conclusion with its provenance. KL requires two evidence items and a previous valid posterior. The agent evaluates the triggers at each eligible iteration.

C. Benchmark Protocol

The 969 review-derived queries are split by review into 656 train, 140 validation, and 173 test queries. The unambiguity filter removes cases whose metadata label disagrees with the relabelled review conclusion. Main test results use 140 Beneficial/NoEffect queries, balanced 70/70. The full filtered test split contains 172 queries, including 32 Harmful cases.

The entity audit covers all 302 distinct intervention/outcome entities in that full set. Retrieved PubMed abstracts span 2010–2025. Every stopping rule is applied to the same recorded per-query evidence sequence, and the available trajectory can end before its maximum budget through source exhaustion.

III. Diagnostics and Robustness

A. Posterior Reliability

Across test trajectories, excluding uncertain steps, approximately 1,800 posterior-confidence/correctness pairs yield ECE 0.236, with overconfidence in high-confidence bins. A high posterior score can therefore overstate the reliability of a conclusion. KL instead tracks the change produced by each new batch, and PRM evaluates graph quality through the learned reward. Calibration, pairwise ranking, and trajectory-level stopping results describe these complementary aspects of the system.

B. Threshold and Path-Discount Sensitivity

The validation sweep covers $\delta \in \{0.001, 0.005, 0.01, 0.05, 0.1\}$ and $\alpha \in \{0.1, 0.2, 0.3, 0.5, 0.7\}$. Across the 25 configurations, accuracy ranges from 63.6% to 65.0% (1.4 percentage points; Fig. 1). PRM decline generally fires before the tested KL thresholds in these trajectories. The narrow range thus reflects the timing of the combined OR triggers: several threshold choices lead the agent to the same stopping states.

The deployed hop discount is $\rho = 1$. On test trajectories, 16/1378 decision steps (1.2%) have a two-hop path within a factor of two of the best direct path. Direct evidence therefore dominates the route portfolio at most decision steps. The discount controls the weight assigned to longer routes, where confidence is compounded across edges.

An exploratory polarity-entropy stopping rule, with threshold 0.5 selected on pooled validation+test data, reaches accuracy comparable to KL-only. This offers another view of belief-based stopping: entropy measures how concentrated the current polarity distribution is, while KL measures how much the distribution changes between rounds. The main comparison uses the validation-frozen KL and PRM thresholds specified above.

C. Grounding and Evidence Exposure

Exact MeSH matches cover 22.5% of the 302 entities (95% CI 17.8–27.2), rising to 43.0% after light normalisation (37.5–48.6). The remaining 57.0% are largely non-atomic clinical phrases; 11 surface forms collapse onto five shared concepts. Across the 172 test queries, both-grounded endpoints occur in 33 cases, with 139 partly or ungrounded cases. Their respective drift rates are 36.4% and 17.3%, with 12.1 and 6.3 abstracts per query (Fig. 2). Greater evidence exposure accompanies greater drift in the grounded group, suggesting that ontology linking and stopping should be evaluated together.

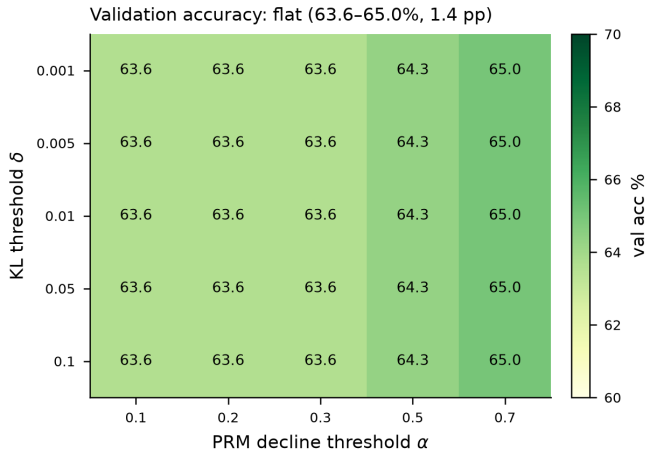


Fig. 1. Validation threshold sweep. Accuracy spans 63.6–65.0% across 25 combinations, with several settings producing the same stopping states under the combined policy.

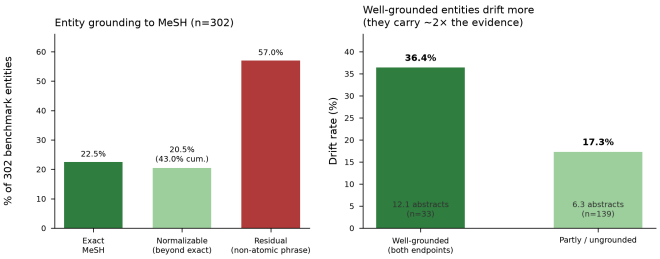


Fig. 2. Grounding audit of the full 172-query test set, including Harmful. Exact and normalisable entities total 43.0% (left); queries with both endpoints grounded have greater evidence exposure and drift (right).

D. The 280-Query Validation and Test Union

On the validation+test union ($n = 280$, balanced 140/140), DACG reaches 66.4% and KL-only 67.5% accuracy (both 95% CI approximately 61–73), compared with 58.9% (53–65) for $k = 20$. NoEffect accuracy is 60.7% for DACG and 59.3% for KL-only, versus 37.9% for the full budget. DACG drift is 8.2% versus 18.6% for $k = 20$, and drift increases across fixed budgets from 7.5% to 18.6%. The larger sample retains the main pattern: adaptive stopping improves null identification and preserves more correct intermediate conclusions than full-budget retrieval.

This analysis combines the validation queries used for threshold selection with the held-out test queries. It describes stopping behaviour across the larger Cochrane sample; the main paper reports the separate 140-query test evaluation.

References

[1] A. B. Tsybakov, *Introduction to Nonparametric Estimation*, ser. Springer Series in Statistics. Springer, 2009.